Web Engineering Course
Final Project

Personal Portfolio Web Application

Assignment 4 Final Project

NA

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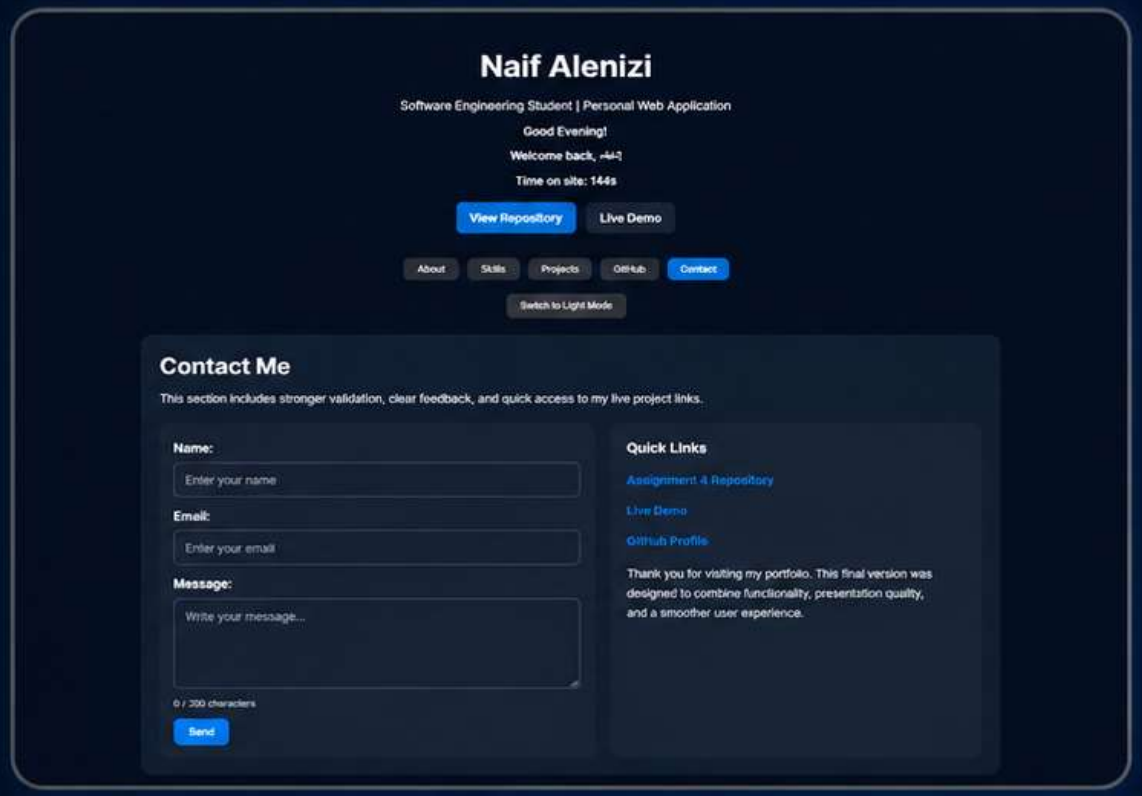
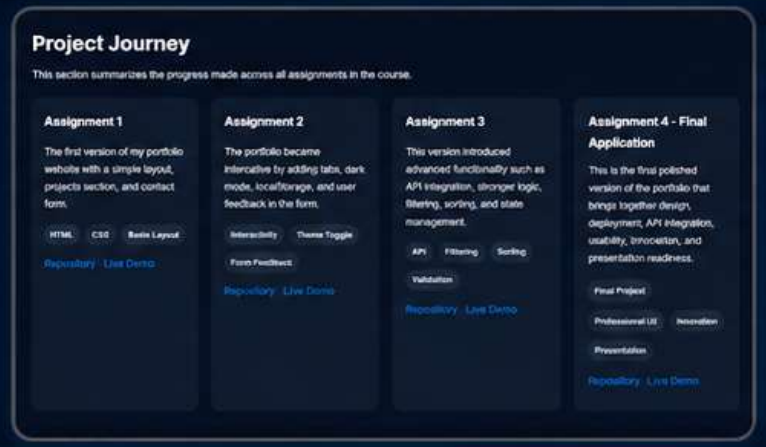
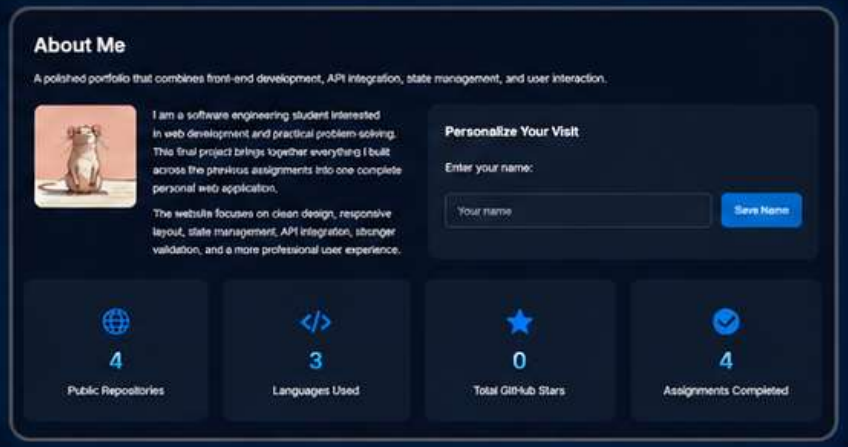
Web Engineering and Development







Project Overview

Final Personal Portfolio Web Application

Project Goal

- Combine previous assignments into one final polished portfolio
- Show stronger front-end development and user interaction
- Demonstrate API integration, state management, and validation
- Present a complete project ready for deployment and demo

Key Features

- Time-based greeting and site timer
- Dark / light mode with saved preference
- Visitor name saving using localStorage
- GitHub repositories with search, filter, sort, and refresh
- Responsive multi-section layout
- Contact form with validation and feedback

About Me

A polished portfolio that combines front-end development, API integration, state management, and user interaction.

I am a software engineering student interested in web development and practical problem-solving. This final project brings together everything I built across the previous assignments into one complete personal web application.

The website focuses on clean design, responsive layout, state management, API integration, stronger validation, and a more professional user experience.

Personalize Your Visit

Enter your name:

Naif | Save Name

4	3	0	4
Public Repositories	Languages Used	Total GitHub Stars	Assignments Completed

Skills & Technical Highlights

A summary of the skills and concepts used to build this final application.

Front-End Development

HTML · CSS · JavaScript · Responsive Design

I used semantic HTML, custom CSS, and JavaScript to build a clean, responsive, and user-friendly interface.

Programming Concepts

API Integration · State Management · Filtering · Sorting · Validation

The project includes dynamic API loading, saved user preferences, search, filtering, sorting, and multi-step form validation.

Tools & Deployment

GitHub · GitHub Pages · Documentation · Presentation Ready

The application is organized for deployment, testing, documentation, and final presentation as a complete project.

Creative Touch

To make the project feel more complete and personal, I added a visitor name feature, quick statistics, a search bar for repositories, and a more polished multi-section interface.

Contact Me

This section includes stronger validation, clear feedback, and quick access to my live project links.

Name:
Enter your name

Email:
Enter your email

Message:
Write your message...

0 / 200 characters

Send

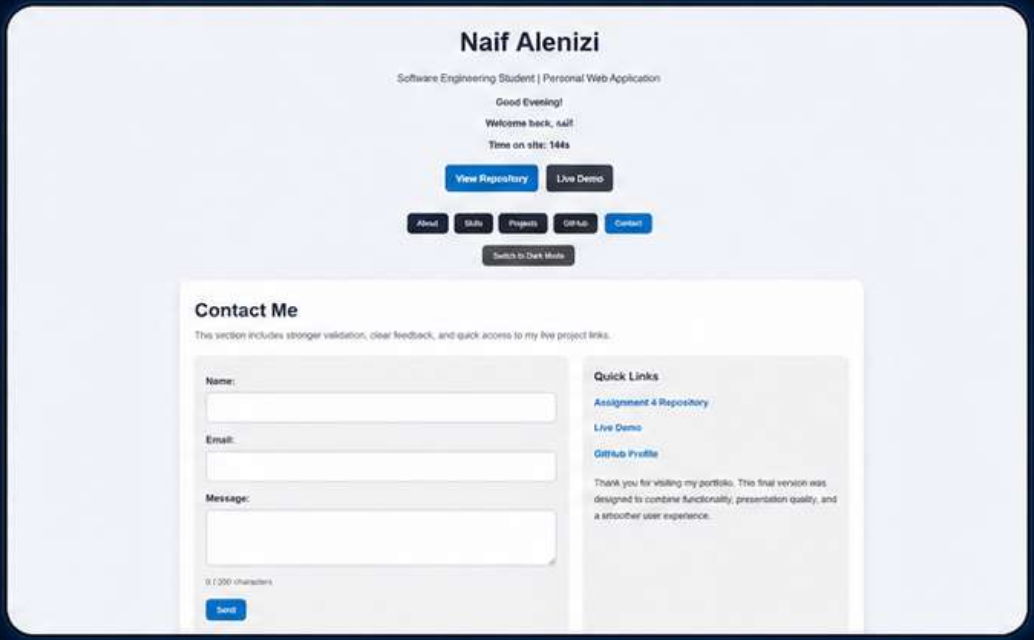
Quick Links

Assignment 4 Repository
Live Demo
GitHub Profile

Thank you for visiting my portfolio. This final version was designed to combine functionality, presentation quality, and a smoother user experience.

Design & Project Structure

How the application is organized



1 Main Sections

- **About** — Overview, background, and introduction.
- **Skills** — Technical skills and competencies.
- **Projects** — Project showcase and details.
- **GitHub** — Links to repositories and activity.
- **Contact** — Contact form and quick links.

2 Technologies Used

- HTML
- CSS
- JavaScript
- GitHub API
- GitHub Pages
- localStorage

A lightweight, frontend-only application using modern web technologies and GitHub integration.

3 Project Files

```
index.html                    # Main HTML structure
css/styles.css                # Styles and responsive design
js/script.js                  # JavaScript functionality
README.md                     # Project overview
docs/ai-usage-report.md       # AI usage report
docs/technical-documentation.md # Technical documentation
```

Application Flow

User Interface
HTML, CSS (Responsive UI)

→

JavaScript Logic
DOM Manipulation, Events, Validation, Routing

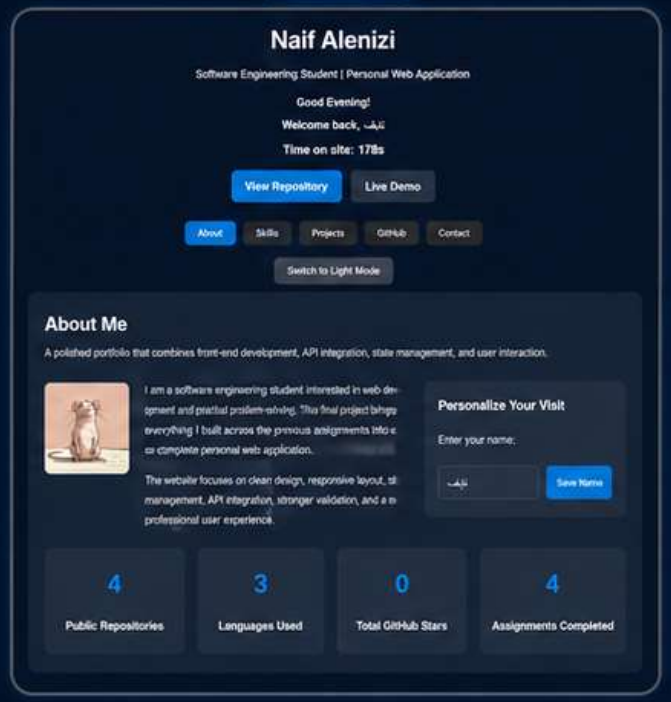
→

GitHub API / localStorage
Fetch Repository Data, Store User Preferences

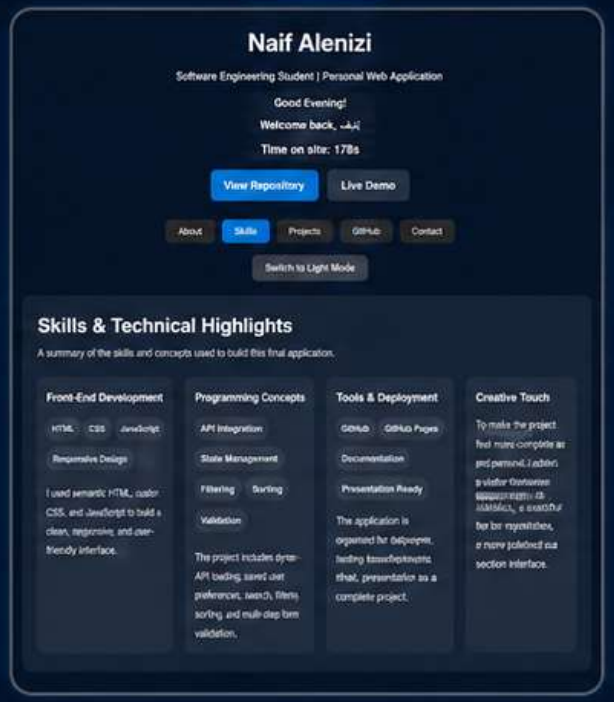
User Interface & Experience

Clean layout, personalization, and polished navigation

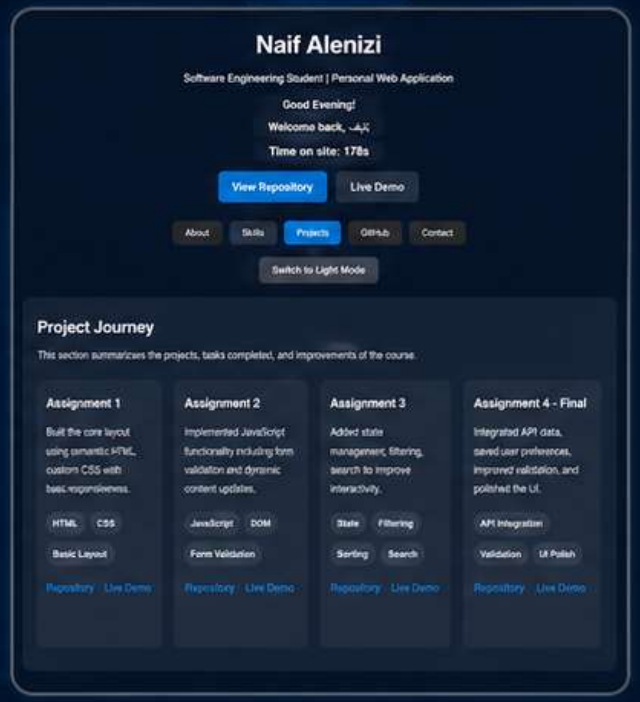
About



Skills



Projects



UX Highlights

- Tabbed navigation between sections
- Dark and light mode toggle
- Saved visitor name and personalization
- Quick GitHub statistics
- Responsive card-based layout
- Clear and simple visual hierarchy

Why it matters

The final version combines usability, presentation quality, and a smoother browsing experience.

GitHub API Integration

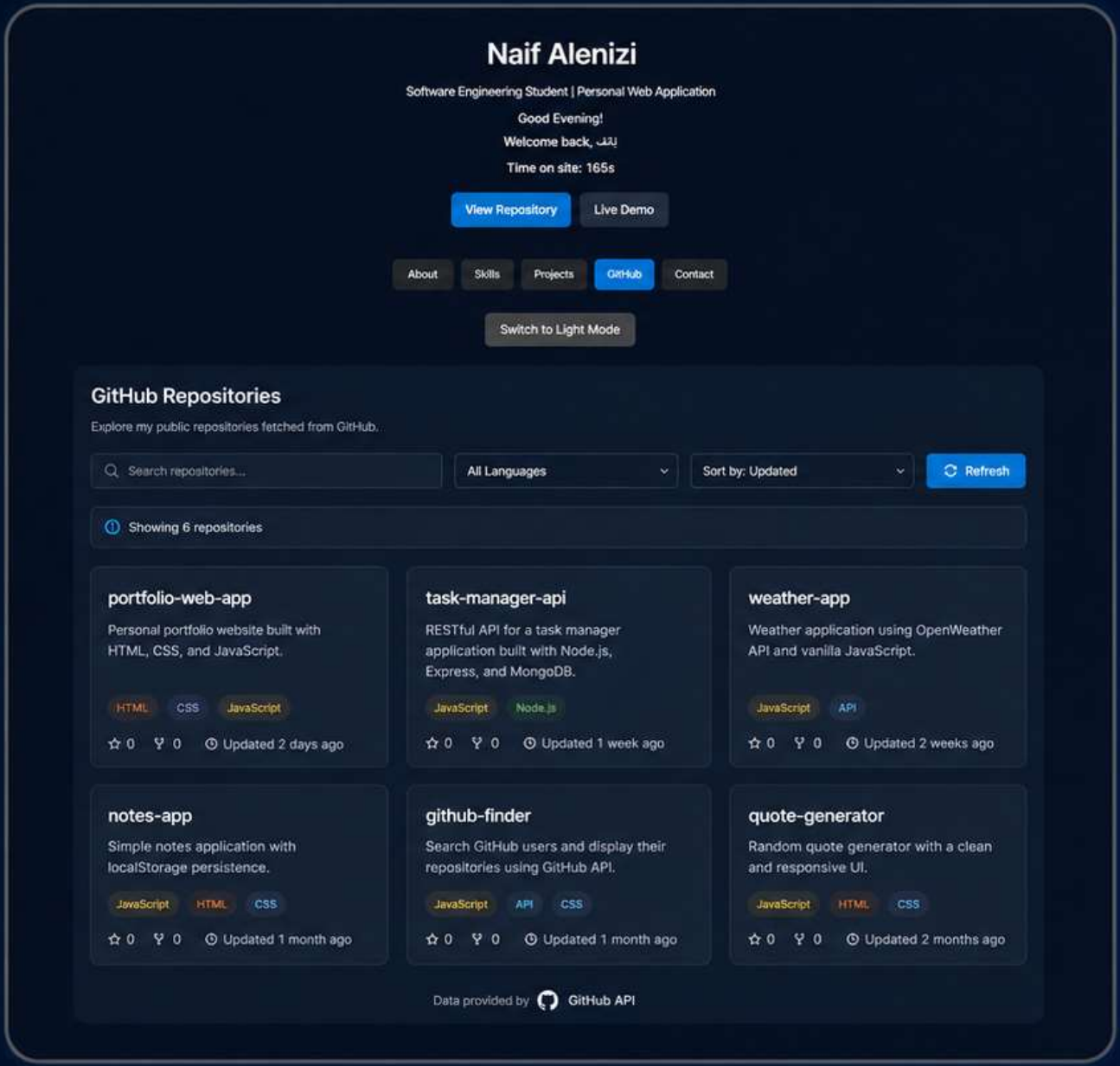
Dynamic data, filtering, and stronger logic

What this section does

- Fetches public repositories from GitHub API
- Displays repository cards dynamically
- Shows language, stars, and updated date
- Supports refresh for live data loading

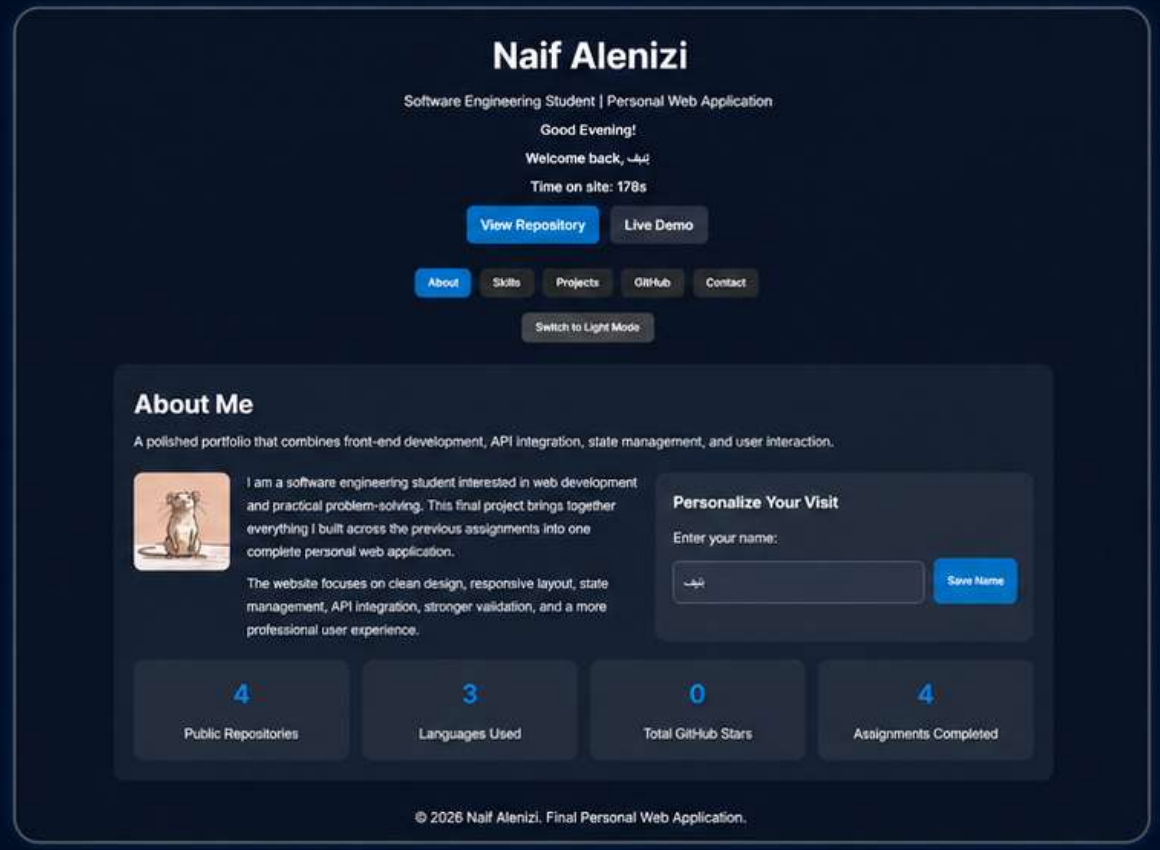
Application Logic

- Search repositories by name or description
- Filter repositories by language
- Sort by last updated, name, or stars
- Show loading, success, error, and empty-state messages
- Save selected filter and sort using localStorage



Validation, Testing & Conclusion

Final checks and project outcome



Form Validation

- Required fields check
- Name length validation
- Valid email format
- Message minimum length
- Character counter and user feedback

Testing Checklist

- Tab navigation works
- Theme preference is saved
- Visitor name persists after refresh
- GitHub API loads correctly
- Filter, sort, and search work
- Links and buttons are functional
- Responsive layout looks good

Conclusion

- Final polished portfolio application
- Combines design, logic, and deployment
- Ready for demonstration and future improvement

Future ideas: backend contact form, more projects, stronger animations